Chapter 5. Potential Problems in Regression

Lecture 6-2 (assumptions, non-Gaussianity, outliers and robust regression, breakdown point, measurement error)

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1 Assumptions and Common Issues

Assumptions: linearity; normal, independent, constant-variance errors; no outliers (large residuals) or high-leverage points (unusual predictors); small collinearity.

Kind of problem	Examples
data problems	outliers, measurement error, missing data
model problems	omitted-variable bias, misspecification, multicollinearity
error problems	non-normality, heteroscedasticity, autocorrelation

2 Non-Gaussianity: Normality Is Not an Issue

Motivation. Estimates are unbiased without normality and the CLT makes inference approximately valid; independence and constant variance matter more.

Theorem: exact distribution under Gaussian errors

If $y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I)$ and X of full column rank, then

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^\top X)^{-1}), \qquad \hat{\beta}_j \sim N(\beta_j, \sigma^2 v_j),$$

where v_j is the jth diagonal element of $(X^\top X)^{-1}$. Replacing σ by $\hat{\sigma} = \sqrt{\mathrm{RSS}/(n-p)}$ (p columns of X including the intercept),

$$t_j = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{v_j}} \sim t_{n-p},$$

which gives tests and confidence intervals.

Proof [Supplement]. $\hat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon$ is a linear map of a Gaussian vector, with mean β and covariance $\sigma^2 (X^\top X)^{-1}$. The residual vector $(I - H)\varepsilon$ is independent of $\hat{\beta}$ (uncorrelated jointly Gaussian), and $\mathrm{RSS}/\sigma^2 = \varepsilon^\top (I - H)\varepsilon / \sigma^2 \sim \chi^2_{n-p}$ since $I - H$ is a projection of rank $n - p$. The ratio of a standard normal and an independent $\sqrt{\chi^2_{n-p}/(n-p)}$ is t_{n-p}. ■

2.1 Large-sample inference

By the CLT,

$$\hat{\beta} \;\dot{\sim}\; N(\beta, \sigma^2 (X^\top X)^{-1}) \quad \text{for large } n,$$

for i.i.d. errors with finite variance and $\max_i h_{ii} \to 0$; t-tests and intervals remain approximately valid. Alternatively, **bootstrap inference** (Ch. 3); resampling pairs (x_i, y_i) is also valid under heteroscedasticity.

3 Outliers and Robust Regression

3.1 Why OLS is sensitive

Squaring residuals in $\sum_{i=1}^n (y_i - x_i^\top \beta)^2$ amplifies large errors, so one extreme observation can dominate the fit.

3.2 Alternative objectives

- **Least absolute deviation (LAD)**: $\hat{\beta} = \arg\min_\beta \sum_i |y_i - x_i^\top \beta|$; targets the conditional median (Exercise 2.1).
- **Huber regression**: $\hat{\beta} = \arg\min_\beta \sum_i \rho_c(r_i)$, $r_i = y_i - x_i^\top \beta$,

$$\rho_c(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le c, \\ c|r| - \frac{1}{2}c^2, & |r| > c. \end{cases}$$

- **Least trimmed squares (LTS)**: $\hat{\beta} = \arg\min_\beta \sum_{i=1}^h r_{(i)}^2$, using only the h smallest squared residuals ($r_{(1)}^2 \le r_{(2)}^2 \le \cdots$).

Derivation: Huber as iteratively reweighted least squares

ρ_c is convex with derivative $\psi_c(r) = r$ for $|r| \le c$ and $c\,\mathrm{sign}(r)$ otherwise. The estimating equation $\sum_i \psi_c(r_i)x_i = 0$ equals the weighted normal equation $\sum_i w_i r_i x_i = 0$ with $w_i = \min(1, c/|r_i|)$, so iterate $\beta \leftarrow (X^\top W X)^{-1} X^\top W y$. Influence is bounded by c (unbounded under OLS). Usually $c = 1.345\hat{s}$, $\hat{s}$ a robust scale (95% efficiency under normality).

3.3 Breakdown point

Definition: breakdown point

The (finite-sample) breakdown point is the smallest fraction of contaminated observations that can make the estimate arbitrarily large.

OLS	LAD	Huber M-estimator	least median of squares	least trimmed squares
0	0	0	0.5	≈ 0.5

OLS: breakdown $1/n \to 0$ (Exercise 5.1). LAD and Huber resist outliers in y but not a bad *leverage point*. LMS (minimize the median squared residual) and LTS with $h \approx n/2$ tolerate nearly 50%, the maximum possible.

4 Measurement Error

True model $Y = \beta_0 + \beta_1 X + \varepsilon$, but

$$\tilde{X} = X + u, \qquad u \text{ independent of } (X, \varepsilon),\ \mathbb{E}u = 0.$$

Regressing Y on $\tilde{X}$:

Theorem: attenuation bias

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 \frac{\mathrm{Var}(X)}{\mathrm{Var}(X) + \mathrm{Var}(u)} = \lambda\beta_1, \qquad 0 < \lambda \le 1.$$

Proof [Supplement]. The OLS slope converges to $\mathrm{Cov}(\tilde{X}, Y)/\mathrm{Var}(\tilde{X})$. Here $\mathrm{Cov}(X + u, \beta_0 + \beta_1 X + \varepsilon) = \beta_1 \mathrm{Var}X$ and $\mathrm{Var}(\tilde{X}) = \mathrm{Var}X + \mathrm{Var}u$. ■

Attenuation bias toward zero; λ is the reliability ratio. Error in Y only inflates the error variance (Exercise 5.2). In multiple regression, error in one predictor also biases correlated predictors' coefficients.

4.1 Remedies

Method	Idea
instrumental variables	Z correlated with X, independent of the error: $\hat\beta_1^{\mathrm{IV}} = \widehat{\mathrm{Cov}}(Z,Y)/\widehat{\mathrm{Cov}}(Z,\tilde X)$
repeated measurements	averaging m replicates gives $\mathrm{Var}u/m$; replicates estimate $\mathrm{Var}u$, and $\hat\beta_1/\hat\lambda$ corrects the bias
errors-in-variables models	model the measurement process (e.g. Deming regression)
SIMEX	add noise of increasing size and extrapolate back to zero error
validation data	calibrate $\tilde X$ on a subset with true X (regression calibration)

Summary

- Normality is often not critical (CLT, bootstrap).
- Robust regression (LAD, Huber, LTS); only high-breakdown methods resist leverage points.
- Predictor error attenuates the slope; IV, replicates, SIMEX or validation data correct it.

5 Selected exercises

Motivation. One outlier breaks OLS; the IV estimator is consistent.

Exercise 5.1 (One outlier breaks OLS)

Let $X \in \mathbb{R}^{n\times q}$ have full column rank q and first column $\mathbf{1}$, with rows $x_i^\top$, and let $H = X(X^\top X)^{-1}X^\top$ with diagonal entries h_{ii}. Replace y_i by $y_i + t$. Show that the new estimate is $\hat\beta(t) = \hat\beta + t(X^\top X)^{-1}x_i$, that $\|\hat\beta(t)\| \to \infty$ as $|t| \to \infty$, and that the fitted value at x_i moves by $t\,h_{ii}$ with $h_{ii} > 0$.

Solution [Supplement]. $\hat\beta(t) = (X^\top X)^{-1}X^\top(y + te_i) = \hat\beta + t(X^\top X)^{-1}x_i$. $u = (X^\top X)^{-1}x_i \neq 0$ since $(X^\top X)^{-1}$ is invertible and $x_i \neq 0$ (first entry 1), so $\|\hat\beta(t)\| \geq |t|\,\|u\| - \|\hat\beta\| \to \infty$. The fitted value at x_i changes by $x_i^\top tu = t\,h_{ii}$, with $h_{ii} = x_i^\top(X^\top X)^{-1}x_i > 0$ by positive definiteness. Hence the breakdown point of OLS is $1/n$ (cf. Exercise 1.3). ■

Exercise 5.2 (Error in Y, and instrumental variables)

Let $(X_i, Y_i, \dots)$ be i.i.d. with finite second moments, $Y = \beta_0 + \beta_1 X + \varepsilon$, $\mathrm{Var}X > 0$, $\mathbb{E}\varepsilon = 0$, $\mathrm{Cov}(X,\varepsilon) = 0$.
(a) If only $\tilde Y = Y + v$ is observed, with $\mathbb{E}v = 0$ and $\mathrm{Cov}(X,v) = 0$, show that the OLS slope of $\tilde Y$ on X is consistent for β_1.
(b) If only $\tilde X = X + u$ is observed (as in the attenuation theorem) and Z satisfies $\mathrm{Cov}(Z,X) \neq 0$, $\mathrm{Cov}(Z,u) = \mathrm{Cov}(Z,\varepsilon) = 0$, show that $\hat\beta_1^{\mathrm{IV}} = \widehat{\mathrm{Cov}}(Z,Y)/\widehat{\mathrm{Cov}}(Z,\tilde X) \xrightarrow{p} \beta_1$.

Solution [Supplement]. By the LLN every sample covariance converges in probability to its population value; ratios converge when the denominator limit is nonzero (continuous mapping).

(a) $\widehat{\mathrm{Cov}}(X,\tilde Y)/\widehat{\mathrm{Var}}(X) \xrightarrow{p} \mathrm{Cov}(X, \beta_0 + \beta_1 X + \varepsilon + v)/\mathrm{Var}X = \beta_1$; the error $\varepsilon + v$ only raises the noise level.

(b) $\mathrm{Cov}(Z,Y) = \beta_1\mathrm{Cov}(Z,X) + \mathrm{Cov}(Z,\varepsilon) = \beta_1\mathrm{Cov}(Z,X)$ and $\mathrm{Cov}(Z,\tilde X) = \mathrm{Cov}(Z,X) + \mathrm{Cov}(Z,u) = \mathrm{Cov}(Z,X) \neq 0$; the ratio is β_1. ■